

labsheet 5

November 10, 2025

1) Predict the output of the following code snippets. If there is an error, correct the code and then write the output.

(a) for i in range(2, 8, 2): print (i * *)

```
[1]: for i in range(2,8,2):  
      print(i*'*')
```

```
**  
****  
*****
```

(b) def greet(name): return “Hello,” + name print(greet(“World”))

```
[2]: def greet(name):  
      return "Hello, " + name  
      print(greet("World"))
```

Hello, World

c) x = input(“Enter a number”) if x % 2 = 0: print(“Even number”) else print(“Odd number”)

```
[3]: x = int(input("Enter a number"))  
      if x % 2==0:  
          print("Even number")  
      else:  
          print("Odd number")
```

Enter a number 3

Odd number

(d) print(“Hello” + 5)

```
[4]: print("Hello" * 5)
```

HelloHelloHelloHelloHello

```
[5]: print("Hello")
```

Hello

(e) total = 0 for i in range(1, 6): if i % 2 != 0: total += i print(total)

```
[9]: total = 0
    for i in range(1, 6):
        if i % 2 != 0:
            total += i
    print(total)
```

9

(f) def check_age(age): if age >= 18: return "Adult" return "Minor" print(check_age(15))

```
[7]: def check_age(age):
    if age >= 18:
        return "Adult"
    return "Minor"
print(check_age(15))
```

Minor

(g) for i in range(1, 10): if i % 2 == 0: continue print(i)

```
[8]: for i in range(1, 10):
    if i % 2 == 0:
        continue
print(i)
```

9

(h) total = 0 for i in range(1, 5) total = total + i print("Sum is:" total)

```
[10]: total = 0
    for i in range(1, 5):
        total = total + i
    print("Sum is: ", total)
```

Sum is: 10

(i) x = 5 y = 10 z = x > y or x + 5 == y print(z)

```
[11]: x = 5
    y = 10
    z = x > y or x + 5 == y
    print(z)
```

True

(j) for i in range(1, 3): for j in range(2): print(i+j)

```
[12]: for i in range(1, 3):
    for j in range(2):
        print(i+j)
```

1

2

2
3

2) Evaluate the following expressions and give an explanation for the result obtained

```
[13]: a=-2**3**2  
      print(a)
```

-512

```
[14]: a=2-5%-3  
      print(a)
```

3

```
[15]: a=-7//4  
      print(a)
```

-2

```
[16]: a=10//3+3**2-4  
      print(a)
```

8

```
[17]: type ((complex(2,3).real is int))
```

```
[17]: bool
```

3) Write a python program to swap the values of two variables a and b without using a third variable.

```
[18]: a=int(input('first variable:'))  
      b=int(input('second variable:'))  
      a=a+b  
      b=a-b  
      a=a-b  
      print(a,b)
```

first variable: 5
second variable: 3

3 5

4) Write a python program that takes two numbers as input, performs addition, subtraction, multiplication, and division (handle division by zero with a message), and prints results using if-else for zero check

```
[19]: a=int(input('first number:'))  
      b=int(input('second number:'))  
      print(a+b)  
      print(a-b)  
      print(a*b)
```

```

print(a/b)
print(a//b)
if b!=0:
    print(a/b)
else:
    print('cannot divide by 0')

```

first number: 25
second number: 5

30
20
125
5.0
5
5.0

5) Implement a program based on flowchart logic: Take a number, use a for loop to sum its digits (e.g., $123 \rightarrow 1 + 2 + 3 = 6$), and if the sum is even, print “Even sum”, else “Odd sum”

```

[20]: a=input('number:')
sum=0
for i in a:
    sum=sum+int(i)
if sum%2==0:
    print("even sum")
else:
    print("odd sum")

```

number: 5

odd sum

6) Write a program with a function print_pattern(rows) that takes an integer ‘rows’ and uses nested for loops to print a triangle pattern of numbers.

```

[21]: def print_pattern(rows):
        for i in range(1,rows+1):
            for j in range(1,i+1):
                print(j,end=' ')
            print()
print_pattern(3)

```

1
1 2
1 2 3

1. Define name=['AA','BB','CC','DD','EE','FF','ZZ','YY','XX','AD', 'KK', 'MM']
 - a. Print length of the list
 - b. Change the name DD to GG and print the new list without correcting the list.

c. Print the following list items: BB, EE, YY, KK using list slicing.

```
[22]: name = ['AA', 'BB', 'CC', 'DD', 'EE', 'FF', 'ZZ', 'YY', 'XX', 'AD', 'KK', 'MM']
print("Length of the list:", len(name))
name[name.index('DD')] = 'GG'
print("Updated list:", name)
print("Selected items:", [name[1], name[4], name[7], name[10]])
```

Length of the list: 12

Updated list: ['AA', 'BB', 'CC', 'GG', 'EE', 'FF', 'ZZ', 'YY', 'XX', 'AD', 'KK', 'MM']

Selected items: ['BB', 'EE', 'YY', 'KK']

```
[24]: # Define the list
num = [12, 23, 2, 4, -6, 67, 45, 30, 21, 3, 5, 4, 9, 14, 10, 22, 89]

# a. Print index of 5
print("Index of 5:", num.index(5))

# b. Print the index of 4 that lies between 5 and 9
sub_list = num[5:10]
if 4 in sub_list:
    print("Index of 4 between index 5 and 9:", sub_list.index(4) + 5)
else:
    print("Value 4 is not found between index 5 and 9.")

# c. Add the value 101 at the end
num.append(101)
print("After appending 101:", num)

# d. Insert 27 in the 7th index
num.insert(7, 27)
print("After inserting 27 at index 7:", num)

# e. Sort the numbers in descending order
num_desc = sorted(num, reverse=True)
print("Sorted in descending order:", num_desc)

# f. Sort the numbers in ascending order
num_asc = sorted(num)
print("Sorted in ascending order:", num_asc)

# g. Remove element 21 and print the new list
num.remove(21)
print("After removing 21:", num)

# h. Remove an element from the 12th index
```

```
del num[12]
print("After deleting element at index 12:", num)

# i. Reverse your list
num.reverse()
print("Reversed list:", num)
```

Index of 5: 10

Value 4 is not found between index 5 and 9.

After appending 101: [12, 23, 2, 4, -6, 67, 45, 30, 21, 3, 5, 4, 9, 14, 10, 22, 89, 101]

After inserting 27 at index 7: [12, 23, 2, 4, -6, 67, 45, 27, 30, 21, 3, 5, 4, 9, 14, 10, 22, 89, 101]

Sorted in descending order: [101, 89, 67, 45, 30, 27, 23, 22, 21, 14, 12, 10, 9, 5, 4, 4, 3, 2, -6]

Sorted in ascending order: [-6, 2, 3, 4, 4, 5, 9, 10, 12, 14, 21, 22, 23, 27, 30, 45, 67, 89, 101]

After removing 21: [12, 23, 2, 4, -6, 67, 45, 27, 30, 3, 5, 4, 9, 14, 10, 22, 89, 101]

After deleting element at index 12: [12, 23, 2, 4, -6, 67, 45, 27, 30, 3, 5, 4, 14, 10, 22, 89, 101]

Reversed list: [101, 89, 22, 10, 14, 4, 5, 3, 30, 27, 45, 67, -6, 4, 2, 23, 12]

3. Write a Python program to sum all the items in the list [23, 34, -3, 55, 34, 12, 2, 6]

```
[29]: numbers = [23, 34, -3, 55, 34, 12, 2, 6]
del sum
total = (sum(numbers))
print("Sum of all items:", total)
```

Sum of all items: 163

4. Write a Python program to multiply all the items in a list [2, 6, 12, 13, -4, 2.5, 8]

```
[30]: numbers = [2, 6, 12, 13, -4, 2.5, 8]
product = 1
for num in numbers:
    product *= num
print("Product of all items:", product)
```

Product of all items: -149760.0

5. Write a Python program to get the largest number from a list [1, 7, 10, 34, 2, 8].

```
[31]: numbers = [1, 7, 10, 34, 2, 8]
largest = max(numbers)
print("Largest number:", largest)
```

Largest number: 34

6. Create the following lists: [0,0,0,0,0,0], [1,3,5,7,9] and [1,2,4,8,16,32,64,128,256] without listing elements explicitly. (Hint – list comprehension – use a for loop in the list.)

```
[32]: # [0, 0, 0, 0, 0, 0, 0]
zeros = [0 for _ in range(7)]

# [1, 3, 5, 7, 9]
odds = [x for x in range(1, 10, 2)]

# [1, 2, 4, 8, 16, 32, 64, 128, 256]
powers_of_two = [2**i for i in range(9)]

print("Zeros:", zeros)
print("Odd numbers:", odds)
print("Powers of 2:", powers_of_two)
```

```
Zeros: [0, 0, 0, 0, 0, 0, 0]
Odd numbers: [1, 3, 5, 7, 9]
Powers of 2: [1, 2, 4, 8, 16, 32, 64, 128, 256]
```

7. Write a python program to replace the last element in a list with another list. Sample data: [1, 3, 5, 7, 9, 10], [2, 4, 6, 8]. Expected Output: [1, 3, 5, 7, 9, 2, 4, 6, 8]

```
[33]: list1 = [1, 3, 5, 7, 9, 10]
list2 = [2, 4, 6, 8]

# Replace last element with list2
list1 = list1[:-1] + list2
print("Merged list:", list1)
```

```
Merged list: [1, 3, 5, 7, 9, 2, 4, 6, 8]
```

8. Write a Python program to move all zero digits to the end of a given list of numbers. Original list: [3, 4, 0, 0, 0, 6, 2, 0, 6, 7, 6, 0, 0, 0, 9, 10, 7, 4, 4, 5, 3, 0, 0, 2, 9, 7, 1] Expected output: [3, 4, 6, 2, 6, 7, 6, 9, 10, 7, 4, 4, 5, 3, 2, 9, 7, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]

```
[34]: # Original list
numbers = [3, 4, 0, 0, 0, 6, 2, 0, 6, 7, 6, 0, 0, 0, 9, 10, 7, 4, 4, 5, 3, 0, 0, 2, 9, 7, 1]

# Move all non-zero elements to the front
non_zeros = [num for num in numbers if num != 0]

# Count how many zeros are in the list
zero_count = numbers.count(0)

# Append zeros at the end
result = non_zeros + [0] * zero_count
```

```
# Print the result
print("List after moving zeros to the end:", result)
```

List after moving zeros to the end: [3, 4, 6, 2, 6, 7, 6, 9, 10, 7, 4, 4, 5, 3, 2, 9, 7, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]

9. Write a Python program to make a list of 50 random integers between 0 and 20
 - a. Print out the number of times each number is repeated in the list. This may be done by creating a new list with the index as the number and the count of each number as elements.
 - b. Find the average of all the numbers in the list.
 - c. Remove all the duplicate values in the list

```
[35]: import random

# Generate a list of 50 random integers between 0 and 20
numbers = [random.randint(0, 20) for _ in range(50)]
print("Original list:\n", numbers)

# a. Count occurrences of each number using a list where index = number, value ↵
    ↵ = count
count_list = [numbers.count(i) for i in range(21)]
print("\nCount of each number (index = number):")
for i, count in enumerate(count_list):
    print(f"{i}: {count}")

# b. Calculate the average of the numbers
average = sum(numbers) / len(numbers)
print("\nAverage of the numbers:", average)

# c. Remove duplicates
unique_numbers = list(set(numbers))
print("\nList with duplicates removed:\n", unique_numbers)
```

Original list:

[16, 2, 0, 8, 13, 5, 12, 17, 0, 1, 1, 9, 3, 13, 4, 15, 7, 3, 8, 5, 7, 12, 1, 2, 17, 5, 10, 8, 13, 6, 15, 13, 12, 19, 7, 4, 10, 10, 14, 18, 18, 15, 14, 2, 3, 9, 2, 13, 8, 7]

Count of each number (index = number):

0: 2
1: 3
2: 4
3: 3
4: 2
5: 3
6: 1
7: 4


```
8: 4
9: 2
10: 3
11: 0
12: 3
13: 5
14: 2
15: 3
16: 1
17: 2
18: 2
19: 1
20: 0
```

Average of the numbers: 8.72

List with duplicates removed:

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17, 18, 19]
```

10. Write a program to accept a word as input and make a list of all the letters in the word.
Avoid duplication of letters.

```
[36]: # Accept a word as input
word = input("Enter a word: ")

# Create a list of unique letters
unique_letters = []
for letter in word:
    if letter not in unique_letters:
        unique_letters.append(letter)

# Print the result
print("List of unique letters:", unique_letters)
```

Enter a word: hello

List of unique letters: ['h', 'e', 'l', 'o']

```
[ ]:
```